EM3

2026-05-30

```r
data_files <- list.files(
  path = "results",
  pattern = "\\.csv$",
  full.names = TRUE
)

data <- data_files %>%
  map_df(~ read_csv(.x, col_types = cols(
    participant = col_character()
  )))

glimpse(data)
```

```
## Rows: 900
## Columns: 14
## $ participant    <chr> "007", "007", "007", "007", "007", "007", "007", "007",~
## $ block_num      <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ block_gender   <chr> "women", "women", "women", "women", "women", "women", "~
## $ trial_num      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, ~
## $ identity       <chr> "AF14NE", "AF05NE", "AF28NE", "AF31NE", "AF28NE", "AF26~
## $ angle          <chr> "HL", "FL", "S", "S", "FL", "FL", "HL", "S", "S", "FL",~
## $ status         <chr> "old", "old", "old", "new", "old", "old", "new", "old",~
## $ correct_answer <chr> "yes", "yes", "yes", "no", "yes", "yes", "no", "yes", "~
## $ response       <chr> "no", "yes", "yes", "no", "yes", "yes", "yes", "yes", "~
## $ correct        <dbl> 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ confidence     <dbl> 6, 9, 9, 6, 6, 7, 4, 9, 9, 9, 9, 8, 7, 9, 9, 9, 9, 9, 7~
## $ response_rt    <dbl> 2.5556922, 2.3054058, 1.9172785, 1.7972861, 3.0750049, ~
## $ rating_rt      <dbl> 2.809481, 3.011354, 2.511487, 2.027741, 2.295047, 2.528~
## $ file_name      <chr> "AF14NEHL.JPG", "AF05NEFL.JPG", "AF28NES.JPG", "AF31NES~
```

```r
unique(data$participant)
```

```
## [1] "007"  "008"  "0606" "1738" "69"
```

```r
#Convert correct and response to numeric
data <- data %>%
  mutate(
    stimulus = ifelse(correct_answer == "yes", 1, 0),  # old = 1
    response_bin = ifelse(response == "yes", 1, 0)     # yes = 1
  )

#prep confidence
data$confidence <- as.numeric(data$confidence)
```

```r
#sanity check (H1)
cor.test(data$confidence, data$correct)
```

```
##
##  Pearson's product-moment correlation
##
## data:  data$confidence and data$correct
## t = 10.305, df = 898, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2654856 0.3824134
## sample estimates:
##       cor
## 0.3251919
```

```r
#Also H1
data$correct <- as.numeric(data$correct)
data$confidence <- as.numeric(data$confidence)

data %>%
  group_by(participant) %>%
  summarise(meta = cor(confidence, correct))
```

```
## # A tibble: 5 x 2
##   participant  meta
##   <chr>       <dbl>
## 1 007         0.283
## 2 008         0.386
## 3 0606        0.450
## 4 1738        0.259
## 5 69          0.220
```

```r
model <- glm(
  correct ~ confidence,
  data = data,
  family = binomial
)

summary(model)
```

```
##
## Call:
## glm(formula = correct ~ confidence, family = binomial, data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.65285    0.22945  -2.845  0.00444 **
## confidence   0.33777    0.03688   9.159  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 875.20  on 899  degrees of freedom
## Residual deviance: 785.27  on 898  degrees of freedom
## AIC: 789.27
##
## Number of Fisher Scoring iterations: 4
```

Betragter korrekthed:

```
total_trials <- nrow(data)

freddie_trials <- sum(ifelse(data$participant == "007" | data$participant == "008", 1, 0))

#overall korrekthed
correct_overall <- sum(ifelse(data$response == data$correct_answer, 1, 0))/total_trials


#korrekthed pr. person
correctness_table <- data %>%
  group_by(participant) %>%
  summarise(
    n_trials = n(),
    n_correct = sum(response == correct_answer, na.rm = TRUE),
    accuracy = mean(response == correct_answer, na.rm = TRUE)
  )

correctness_table
```

```
## # A tibble: 5 x 4
##   participant n_trials n_correct accuracy
##   <chr>          <int>     <int>    <dbl>
## 1 007              180       149    0.828
## 2 008              180       155    0.861
## 3 0606             180       143    0.794
## 4 1738             180       147    0.817
## 5 69               180       135    0.75
```

```
#Frederikkes korrekthed
correct_EEGFP <- sum(ifelse((data$participant == "007" | data$participant == "008") & data$response == c


#overall korrekthed for kvinde-blokke
correct_overall_women <-
  sum(ifelse(data$block_gender == "women" & data$response == data$correct_answer, 1, 0))/(total_trials/


#overall korrekthed for mande-blokke
correct_overall_men <-
  sum(ifelse(data$block_gender == "men" & data$response == data$correct_answer, 1, 0))/(total_trials/2)


#Frederikke korrekthed for kvinde-blokke
```

```
correct_women_fred <-
  sum(ifelse((data$participant == "007" | data$participant == "008") & data$block_gender == "women" & da

#Frederikke korrekthed for mande-blokke
correct_men_fred <-
  sum(ifelse((data$participant == "007" | data$participant == "008") & data$block_gender == "men" & data
```

```
accuracy_table <- data.frame(
  category = c(
    "Overall",
    "Overall women blocks",
    "Overall men blocks",
    "EEG participant",
    "EEG women blocks",
    "EEG men blocks"
  ),

  accuracy = c(
    correct_overall,
    correct_overall_women,
    correct_overall_men,
    correct_EEGFP,
    correct_women_fred,
    correct_men_fred
  )
)

accuracy_table
```

```
##               category  accuracy
## 1              Overall 0.8100000
## 2 Overall women blocks 0.8222222
## 3   Overall men blocks 0.7977778
## 4      EEG participant 0.8444444
## 5     EEG women blocks 0.8500000
## 6       EEG men blocks 0.8388889
```

Metakognitiv effektivitet

```
library(dplyr)
library(statConfR)

#metakognitiv effektivitet pr. FP

df_meta_indi <- data %>%
  mutate(
    participant = as.integer(participant),
    stimulus = factor(correct_answer, levels = c("no", "yes")),
    response = factor(response, levels = c("no", "yes")),
    correct = as.numeric(correct),
    rating = as.factor(confidence)
  ) %>%
```

```r
  select(participant, stimulus, response, correct, rating)

meta_results <- fitMetaDprime(
  data = df_meta_indi,
  model = "ML"
)

meta_results
```

```
##   model participant   dprime           c     metaD     Ratio
## 1    ML           7 2.168294  0.56605738 0.9222652 0.4253413
## 2    ML           8 2.256872  0.33679759 1.7766476 0.7872167
## 3    ML         606 1.621542  0.01913239 2.1852962 1.3476655
## 4    ML        1738 1.788810  0.10276640 1.3275725 0.7421540
## 5    ML          69 1.378559 -0.26358385 1.0047329 0.7288284
```

```r
#global metakognitiv effektivitet
fred_meta <- mean(meta_results$Ratio)
```

Tjekker om metakognitiv sensitvitet er bedre end chance (H1)

```r
t.test(
  meta_results$metaD,
  mu = 0,
  alternative = "greater"
)
```

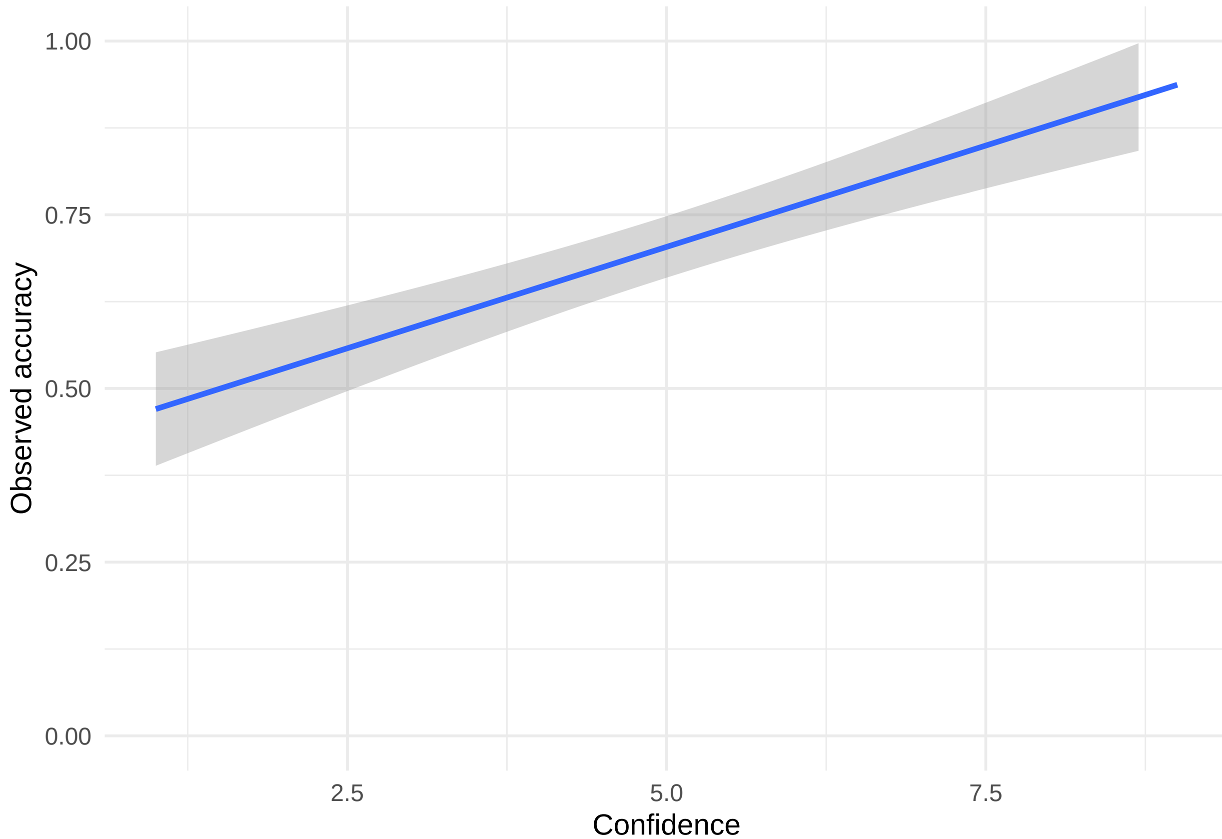
```
##
##  One Sample t-test
##
## data:  meta_results$metaD
## t = 6.0483, df = 4, p-value = 0.001885
## alternative hypothesis: true mean is greater than 0
## 95 percent confidence interval:
##  0.9345793       Inf
## sample estimates:
## mean of x
##  1.443303
```

Plotter

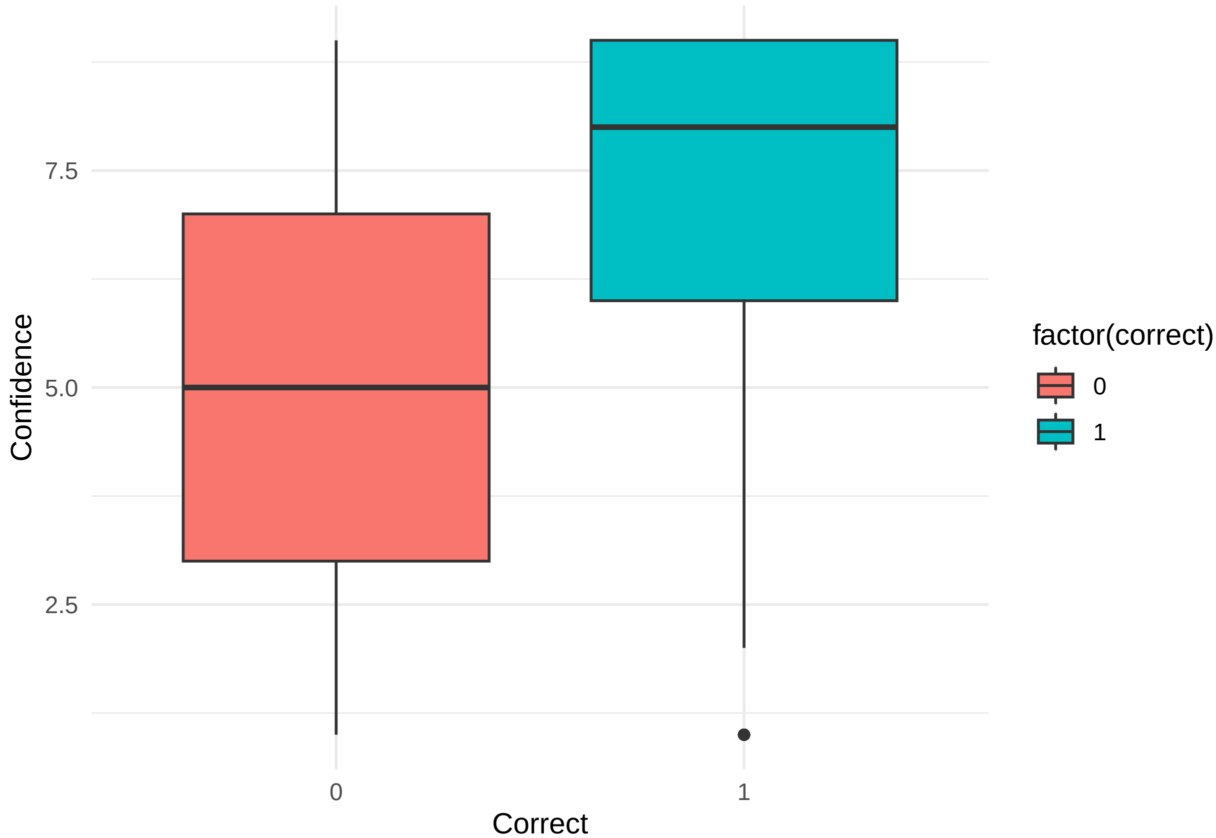
```r
calibration <- data %>%
  group_by(confidence) %>%
  summarise(
    accuracy = mean(correct),
    n = n()
  )

ggplot(calibration,
       aes(x = confidence,
           y = accuracy)) +
  geom_smooth(
```

```r
    method = "lm",
    se = TRUE
  ) +
  ylim(0,1) +
  labs(
    x = "Confidence",
    y = "Observed accuracy"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
#H1

ggplot(data,
       aes(x = factor(correct),
           y = confidence,
           fill = factor(correct))) +
  geom_boxplot() +
  labs(
    x = "Correct",
    y = "Confidence"
  ) +
  theme_minimal()
```




```
#Prep for EEG comparison
data_eeg <- data %>%
  filter(participant %in% c("007", "008"))

data_eeg <- data_eeg %>%
  arrange(participant, block_num, trial_num)

glimpse(data_eeg)
```

```
## Rows: 360
## Columns: 16
## $ participant    <chr> "007", "007", "007", "007", "007", "007", "007", "007",~
## $ block_num      <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ block_gender   <chr> "women", "women", "women", "women", "women", "women", "~
## $ trial_num      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, ~
## $ identity       <chr> "AF14NE", "AF05NE", "AF28NE", "AF31NE", "AF28NE", "AF26~
## $ angle          <chr> "HL", "FL", "S", "S", "FL", "FL", "HL", "S", "S", "FL",~
## $ status         <chr> "old", "old", "old", "new", "old", "old", "new", "old",~
## $ correct_answer <chr> "yes", "yes", "yes", "no", "yes", "yes", "no", "yes", "~
## $ response       <chr> "no", "yes", "yes", "no", "yes", "yes", "yes", "yes", "~
## $ correct        <dbl> 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ confidence     <dbl> 6, 9, 9, 6, 6, 7, 4, 9, 9, 9, 9, 8, 7, 9, 9, 9, 9, 9, 7~
## $ response_rt    <dbl> 2.5556922, 2.3054058, 1.9172785, 1.7972861, 3.0750049, ~
## $ rating_rt      <dbl> 2.809481, 3.011354, 2.511487, 2.027741, 2.295047, 2.528~
## $ file_name      <chr> "AF14NEHL.JPG", "AF05NEFL.JPG", "AF28NES.JPG", "AF31NES~
## $ stimulus       <dbl> 1, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 1~
```

```
## $ response_bin   <dbl> 0, 1, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 1~
```

Now EEG:

```r
eeg <- read_csv("eeg_features.csv")
```

```
## Rows: 360 Columns: 16
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (7): block_gender, identity, angle, status, correct_answer, response, fi...
## dbl (9): participant, block_num, trial_num, correct, confidence, response_rt...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
eeg$participant <- sprintf("%03d", eeg$participant)

nrow(data_eeg)
```

```
## [1] 360
```

```r
nrow(eeg)
```

```
## [1] 360
```

Now, merge:

```r
full_data <- left_join(
  data_eeg,
  eeg,
  by = c("participant", "block_num", "trial_num")
)

glimpse(full_data)
```

```
## Rows: 360
## Columns: 29
## $ participant      <chr> "007", "007", "007", "007", "007", "007", "007", "007~
## $ block_num        <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ block_gender.x   <chr> "women", "women", "women", "women", "women", "women",~
## $ trial_num        <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16~
## $ identity.x       <chr> "AF14NE", "AF05NE", "AF28NE", "AF31NE", "AF28NE", "AF~
## $ angle.x          <chr> "HL", "FL", "S", "S", "FL", "FL", "HL", "S", "S", "FL~
## $ status.x         <chr> "old", "old", "old", "new", "old", "old", "new", "old~
## $ correct_answer.x <chr> "yes", "yes", "yes", "no", "yes", "yes", "no", "yes",~
## $ response.x       <chr> "no", "yes", "yes", "no", "yes", "yes", "yes", "yes",~
## $ correct.x        <dbl> 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ confidence.x     <dbl> 6, 9, 9, 6, 6, 7, 4, 9, 9, 9, 9, 8, 7, 9, 9, 9, 9, 9,~
## $ response_rt.x    <dbl> 2.5556922, 2.3054058, 1.9172785, 1.7972861, 3.0750049~
## $ rating_rt.x      <dbl> 2.809481, 3.011354, 2.511487, 2.027741, 2.295047, 2.5~
## $ file_name.x      <chr> "AF14NEHL.JPG", "AF05NEFL.JPG", "AF28NES.JPG", "AF31N~
```

```
## $ stimulus         <dbl> 1, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1,~
## $ response_bin     <dbl> 0, 1, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1,~
## $ block_gender.y   <chr> "women", "women", "women", "women", "women", "women",~
## $ identity.y       <chr> "AF14NE", "AF05NE", "AF28NE", "AF31NE", "AF28NE", "AF~
## $ angle.y          <chr> "HL", "FL", "S", "S", "FL", "FL", "HL", "S", "S", "FL~
## $ status.y         <chr> "old", "old", "old", "new", "old", "old", "new", "old~
## $ correct_answer.y <chr> "yes", "yes", "yes", "no", "yes", "yes", "no", "yes",~
## $ response.y       <chr> "no", "yes", "yes", "no", "yes", "yes", "yes", "yes",~
## $ correct.y        <dbl> 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ confidence.y     <dbl> 6, 9, 9, 6, 6, 7, 4, 9, 9, 9, 9, 8, 7, 9, 9, 9, 9, 9,~
## $ response_rt.y    <dbl> 2.5556922, 2.3054058, 1.9172785, 1.7972861, 3.0750049~
## $ rating_rt.y      <dbl> 2.809481, 3.011354, 2.511487, 2.027741, 2.295047, 2.5~
## $ file_name.y      <chr> "AF14NEHL.JPG", "AF05NEFL.JPG", "AF28NES.JPG", "AF31N~
## $ theta_power      <dbl> 8.614230e-10, 8.102975e-10, 1.114902e-09, 4.285158e-1~
## $ alpha_power      <dbl> 2.149729e-10, 1.210692e-10, 2.207116e-10, 1.624044e-1~
```

```r
#deskriptiv figurer:

library(dplyr)
library(tidyr)

summary_table <- data %>%
  mutate(
    picture = ifelse(correct_answer == "yes", "Old picture", "New picture"),
    response_type = ifelse(response == "yes", "Old response", "New response"),

    confidence_group = case_when(
      confidence <= 3 ~ "Low confidence",
      confidence <= 6 ~ "Medium confidence",
      TRUE ~ "High confidence"
    ),

    confidence_group = factor(
      confidence_group,
      levels = c(
        "Low confidence",
        "Medium confidence",
        "High confidence"
      )
    ),

    eeg_participant = participant %in% c("007", "008")
  ) %>%

  group_by(picture, response_type, confidence_group) %>%
  summarise(
    total_count = n(),
    eeg_count = sum(eeg_participant),
    .groups = "drop"
  )

# summeringer for confidence-grupper
totals <- summary_table %>%
  group_by(confidence_group) %>%
```

```
  summarise(
    total_count = sum(total_count),
    eeg_count = sum(eeg_count),
    .groups = "drop"
  ) %>%
  mutate(
    picture = "TOTAL",
    response_type = ""
  )

# combine
summary_table <- bind_rows(summary_table, totals) %>%

  mutate(
    display = paste0(total_count, " (", eeg_count, ")")
  ) %>%

  select(picture, response_type, confidence_group, display) %>%

  pivot_wider(
    names_from = confidence_group,
    values_from = display
  )

summary_table
```

```
## # A tibble: 5 x 5
##   picture   response_type `Low confidence` `Medium confidence` `High confidence`
##   <chr>     <chr>         <chr>            <chr>               <chr>
## 1 New pict~ "New respons~ 49 (6)           83 (25)             246 (139)
## 2 New pict~ "Old respons~ 23 (4)           28 (5)              21 (1)
## 3 Old pict~ "New respons~ 23 (8)           40 (11)             36 (27)
## 4 Old pict~ "Old respons~ 29 (9)           58 (17)             264 (108)
## 5 TOTAL     ""            124 (27)         209 (58)            567 (275)
```

Now lets model this shit

```
#Theta !!!
model_theta <- glm(
  correct.x ~ confidence.x * theta_power,
  data = full_data,
  family = binomial
)

summary(model_theta)
```

```
##
## Call:
## glm(formula = correct.x ~ confidence.x * theta_power, family = binomial,
##     data = full_data)
##
## Coefficients:
```

```
##                              Estimate Std. Error z value Pr(>|z|)
## (Intercept)                 6.884e-01  1.363e+00   0.505    0.614
## confidence.x                9.842e-02  1.944e-01   0.506    0.613
## theta_power                -2.277e+09  1.938e+09  -1.174    0.240
## confidence.x:theta_power  3.892e+08  2.790e+08   1.395    0.163
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 311.20  on 359  degrees of freedom
## Residual deviance: 275.52  on 356  degrees of freedom
## AIC: 283.52
##
## Number of Fisher Scoring iterations: 5
```

Alpha:

```
model_alpha <- glm(
  correct.x ~ confidence.x * alpha_power,
  data = full_data,
  family = binomial
)

summary(model_alpha)
```

```
##
## Call:
## glm(formula = correct.x ~ confidence.x * alpha_power, family = binomial,
##     data = full_data)
##
## Coefficients:
##                              Estimate Std. Error z value Pr(>|z|)
## (Intercept)                -2.018e+00  1.758e+00  -1.148    0.251
## confidence.x                3.833e-01  2.480e-01   1.546    0.122
## alpha_power                 1.004e+10  1.447e+10   0.693    0.488
## confidence.x:alpha_power -2.624e+08  2.035e+09  -0.129    0.897
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 311.20  on 359  degrees of freedom
## Residual deviance: 274.53  on 356  degrees of freedom
## AIC: 282.53
##
## Number of Fisher Scoring iterations: 5
```

Clean/scaled Theta (?)

```
full_data <- full_data %>%
  mutate(theta_z = as.numeric(scale(theta_power)))

model_thetaz <- glm(
  correct.x ~ confidence.x * theta_z + factor(block_num),
  data = full_data,
  family = binomial
```

```r
)

summary(model_thetaz)
```

```
##
## Call:
## glm(formula = correct.x ~ confidence.x * theta_z + factor(block_num),
##     family = binomial, data = full_data)
##
## Coefficients:
##                       Estimate Std. Error z value Pr(>|z|)
## (Intercept)           -0.52783    0.59527  -0.887    0.375
## confidence.x           0.36907    0.06559   5.627 1.84e-08 ***
## theta_z               -0.61990    0.51001  -1.215    0.224
## factor(block_num)2    -0.44689    0.57861  -0.772    0.440
## factor(block_num)3     0.01157    0.60204   0.019    0.985
## factor(block_num)4    -0.48516    0.55215  -0.879    0.380
## factor(block_num)5    -0.64972    0.54106  -1.201    0.230
## factor(block_num)6    -0.49332    0.55850  -0.883    0.377
## confidence.x:theta_z   0.10607    0.07333   1.446    0.148
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 311.20  on 359  degrees of freedom
## Residual deviance: 272.94  on 351  degrees of freedom
## AIC: 290.94
##
## Number of Fisher Scoring iterations: 5
```

Same for Alpha:

```r
full_data <- full_data %>%
  mutate(alpha_z = as.numeric(scale(alpha_power)))

model_alphaz <- glm(
  correct.x ~ confidence.x * alpha_z + factor(block_num),
  data = full_data,
  family = binomial
)

summary(model_alphaz)
```

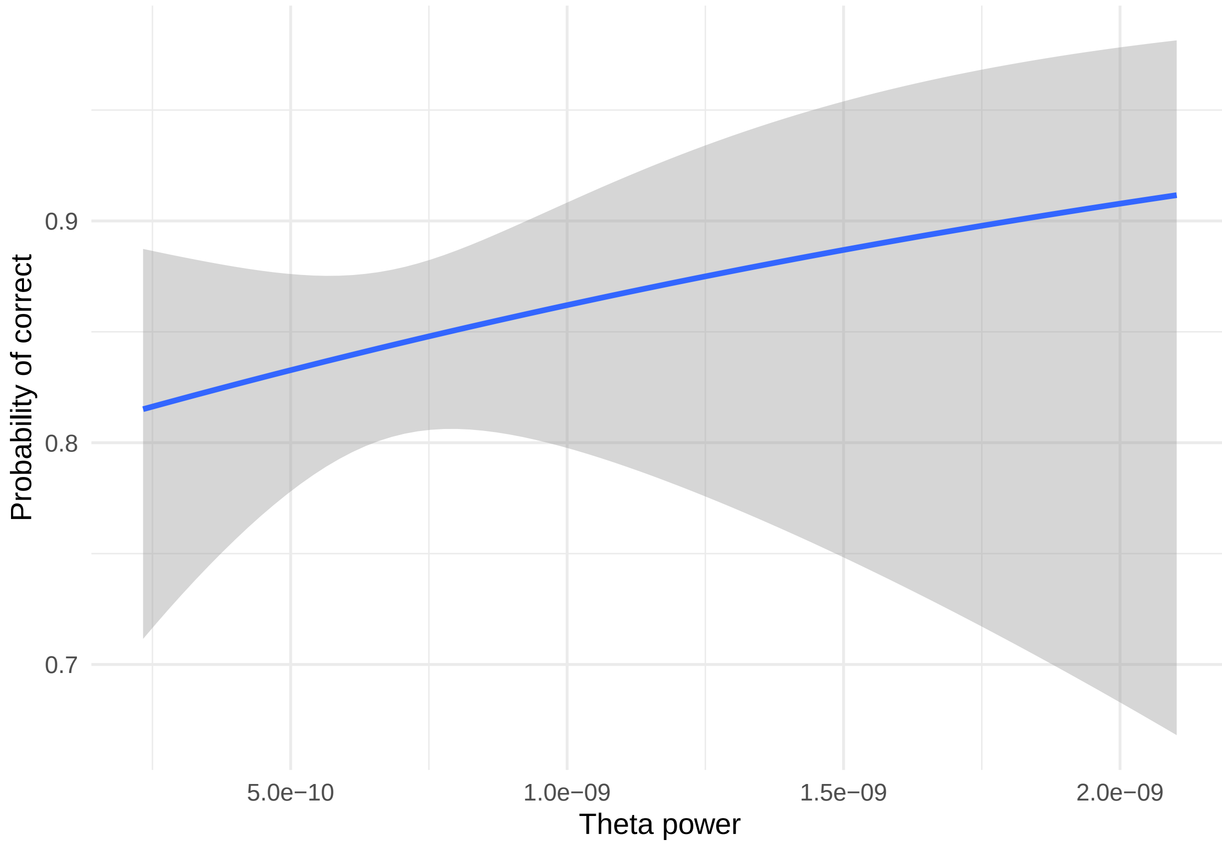
```
##
## Call:
## glm(formula = correct.x ~ confidence.x * alpha_z + factor(block_num),
##     family = binomial, data = full_data)
##
## Coefficients:
##                         Estimate Std. Error z value Pr(>|z|)
## (Intercept)           -0.366896   0.605832  -0.606    0.545
```

```
## confidence.x          0.348731   0.067517   5.165  2.4e-07 ***
## alpha_z               0.318234   0.537835   0.592    0.554
## factor(block_num)2   -0.396351   0.580558  -0.683    0.495
## factor(block_num)3   -0.017531   0.598294  -0.029    0.977
## factor(block_num)4   -0.609824   0.552958  -1.103    0.270
## factor(block_num)5   -0.583126   0.553918  -1.053    0.292
## factor(block_num)6   -0.439311   0.560611  -0.784    0.433
## confidence.x:alpha_z -0.002107   0.075174  -0.028    0.978
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 311.20  on 359  degrees of freedom
## Residual deviance: 272.13  on 351  degrees of freedom
## AIC: 290.13
##
## Number of Fisher Scoring iterations: 5
```

Lets make some plots (Theta)

```
ggplot(full_data, aes(x = theta_power, y = correct.x))+
  geom_smooth(method = "glm", method.args = list(family = "binomial")) +
  labs(
    x = "Theta power",
    y = "Probability of correct"
  ) +
  theme_minimal()
```

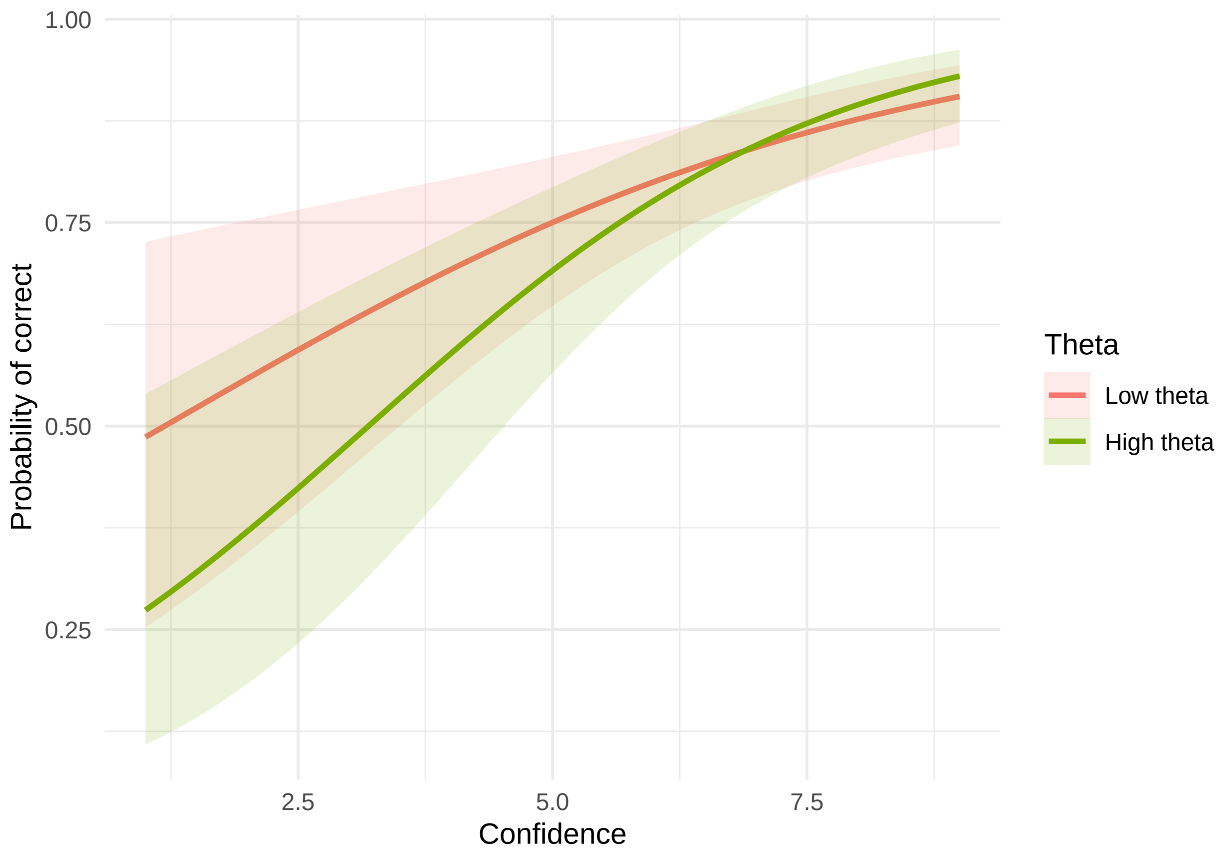
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
full_data$theta_group <- cut(
  full_data$theta_z,
  breaks = c(-Inf, 0, Inf),
  labels = c("Low theta", "High theta")
)

ggplot(full_data,
       aes(x = confidence.x,
           y = correct.x,
           color = theta_group,
           fill = theta_group)) +

  geom_smooth(
    method = "glm",
    method.args = list(family = "binomial"),
    se = TRUE,
    alpha = 0.15
  ) +

  scale_color_manual(values = c(
    "Low theta" = "#F8766D",
    "High theta" = "#7CAE00"
  )) +

  scale_fill_manual(values = c(
    "Low theta" = "#F8766D",
```

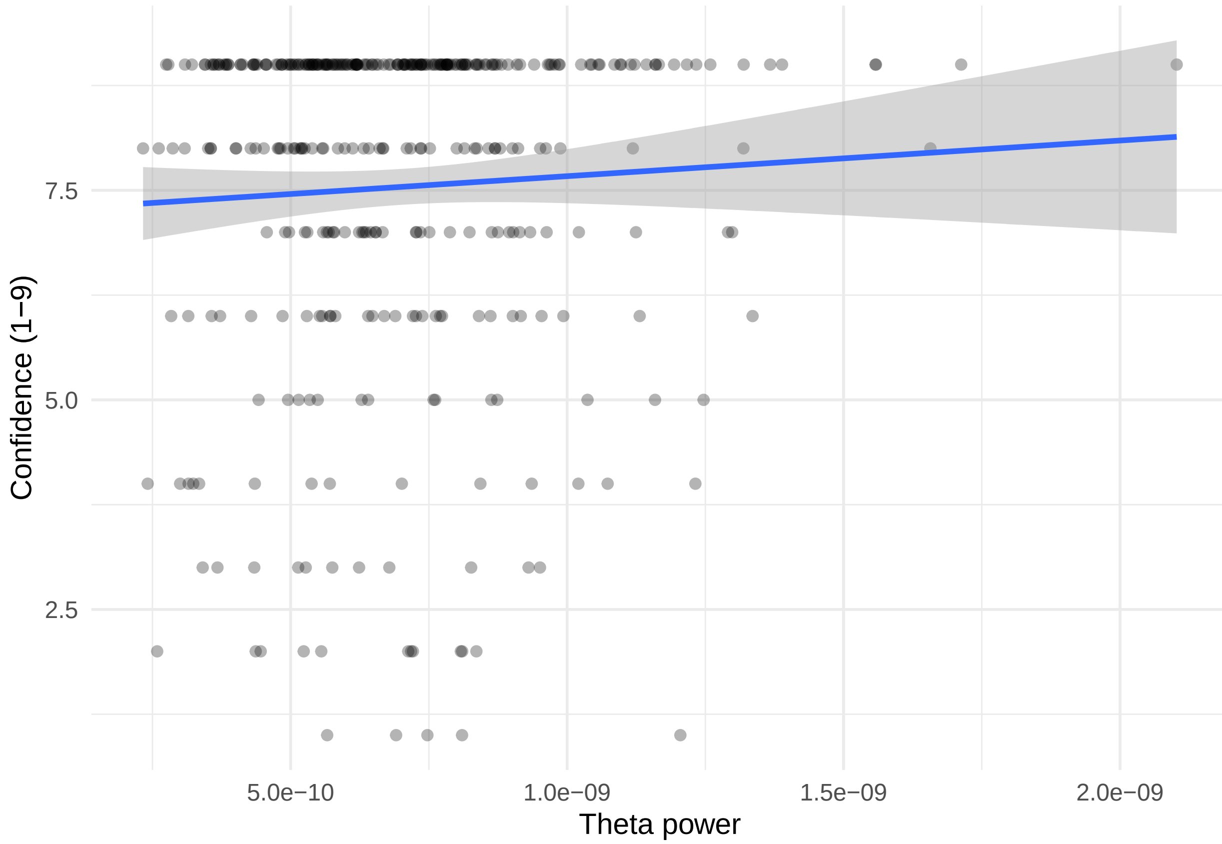
```r
    "High theta" = "#7CAE00"
  )) +

  labs(
    x = "Confidence",
    y = "Probability of correct",
    color = "Theta",
    fill = "Theta"
  ) +

  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
ggplot(full_data, aes(x = theta_power, y = confidence.x)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm") +
  labs(
    x = "Theta power",
    y = "Confidence (1-9)"
  ) +
  theme_minimal()
```

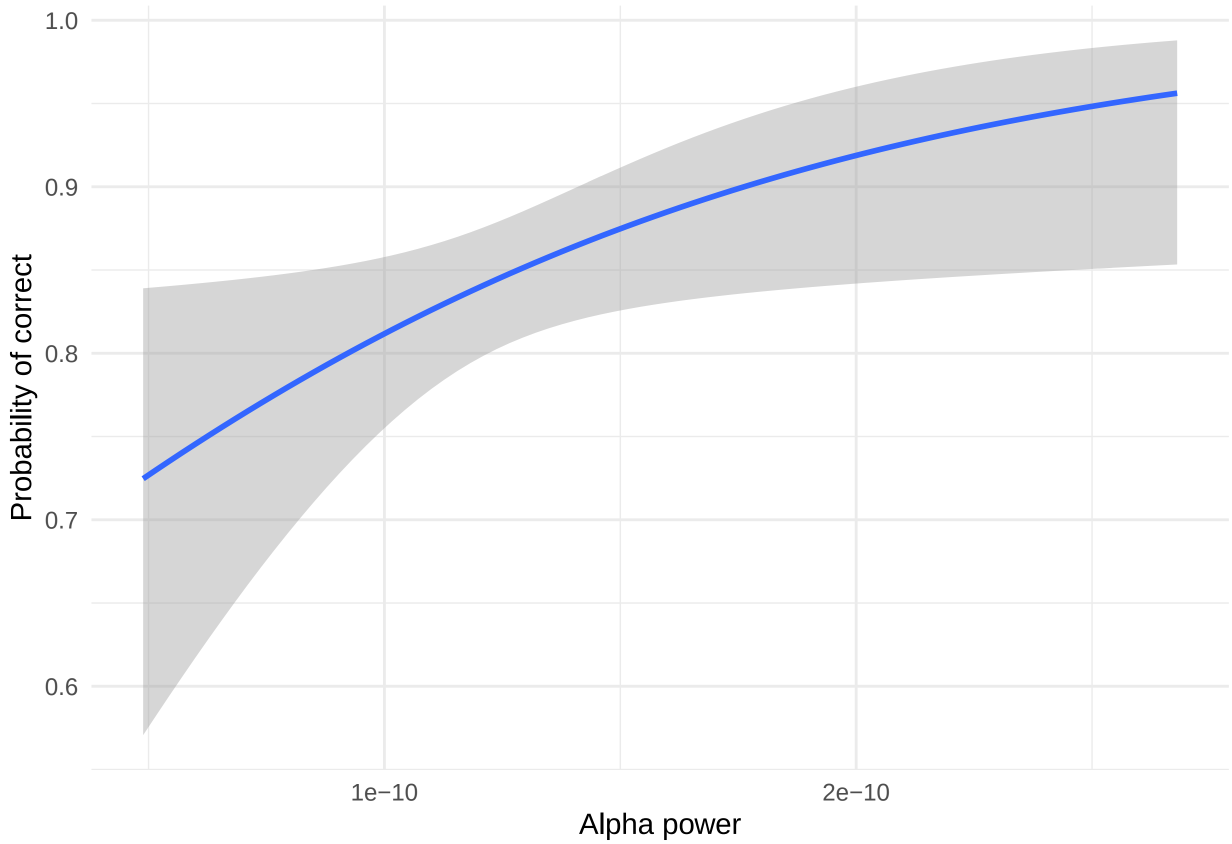
```
## `geom_smooth()` using formula = 'y ~ x'
```



Plot med alpha

```
ggplot(full_data, aes(x = alpha_power, y = correct.x))+
  geom_smooth(method = "glm", method.args = list(family = "binomial")) +
  labs(
    x = "Alpha power",
    y = "Probability of correct"
  ) +
  theme_minimal()
```

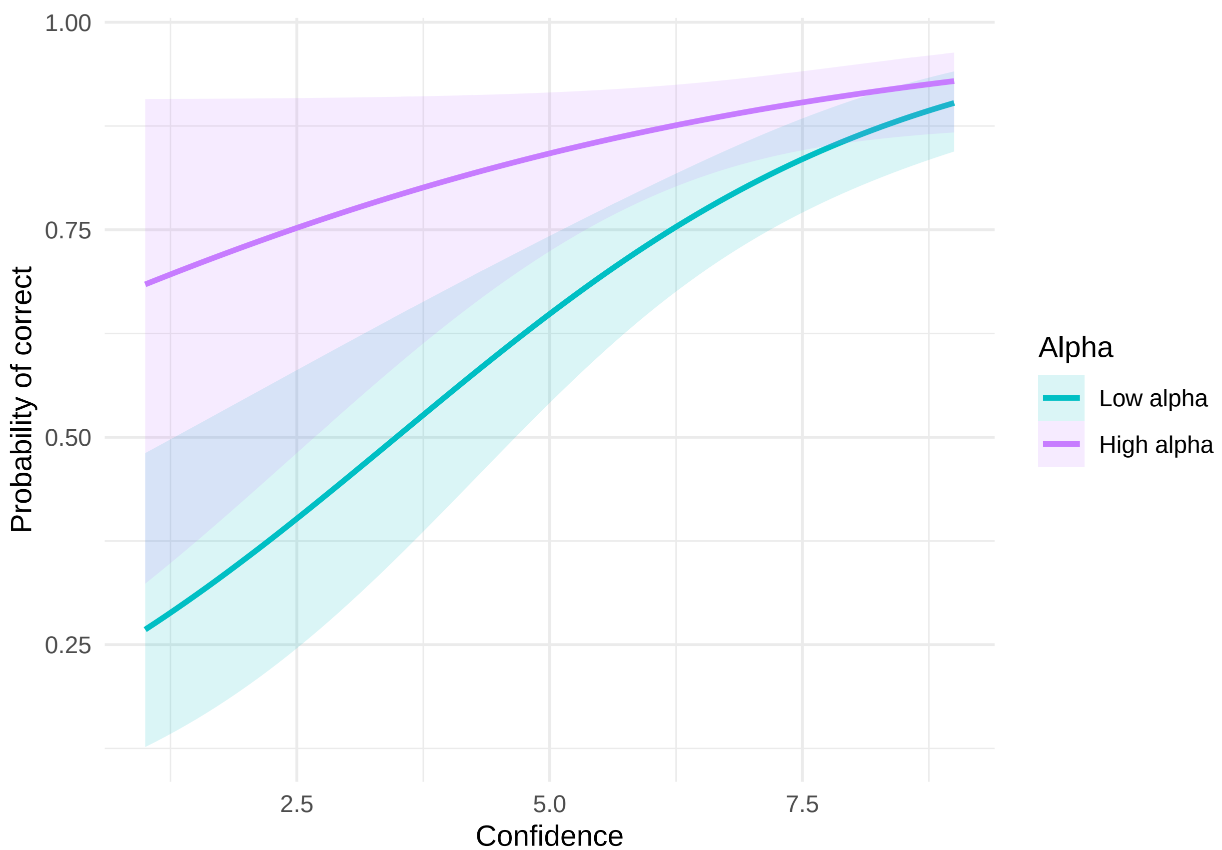
```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
full_data$alpha_group <- cut(
  full_data$alpha_z,
  breaks = c(-Inf, 0, Inf),
  labels = c("Low alpha", "High alpha")
)

ggplot(full_data,
       aes(x = confidence.x,
           y = correct.x,
           color = alpha_group,
           fill = alpha_group)) +

  geom_smooth(
    method = "glm",
    method.args = list(family = "binomial"),
    se = TRUE,
    alpha = 0.15
  ) +

  scale_color_manual(values = c(
    "Low alpha" = "#00BFC4",
    "High alpha" = "#C77CFF"
  )) +

  scale_fill_manual(values = c(
    "Low alpha" = "#00BFC4",
```

```
    "High alpha" = "#C77CFF"
  )) +

  labs(
    x = "Confidence",
    y = "Probability of correct",
    color = "Alpha",
    fill = "Alpha"
  ) +

  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



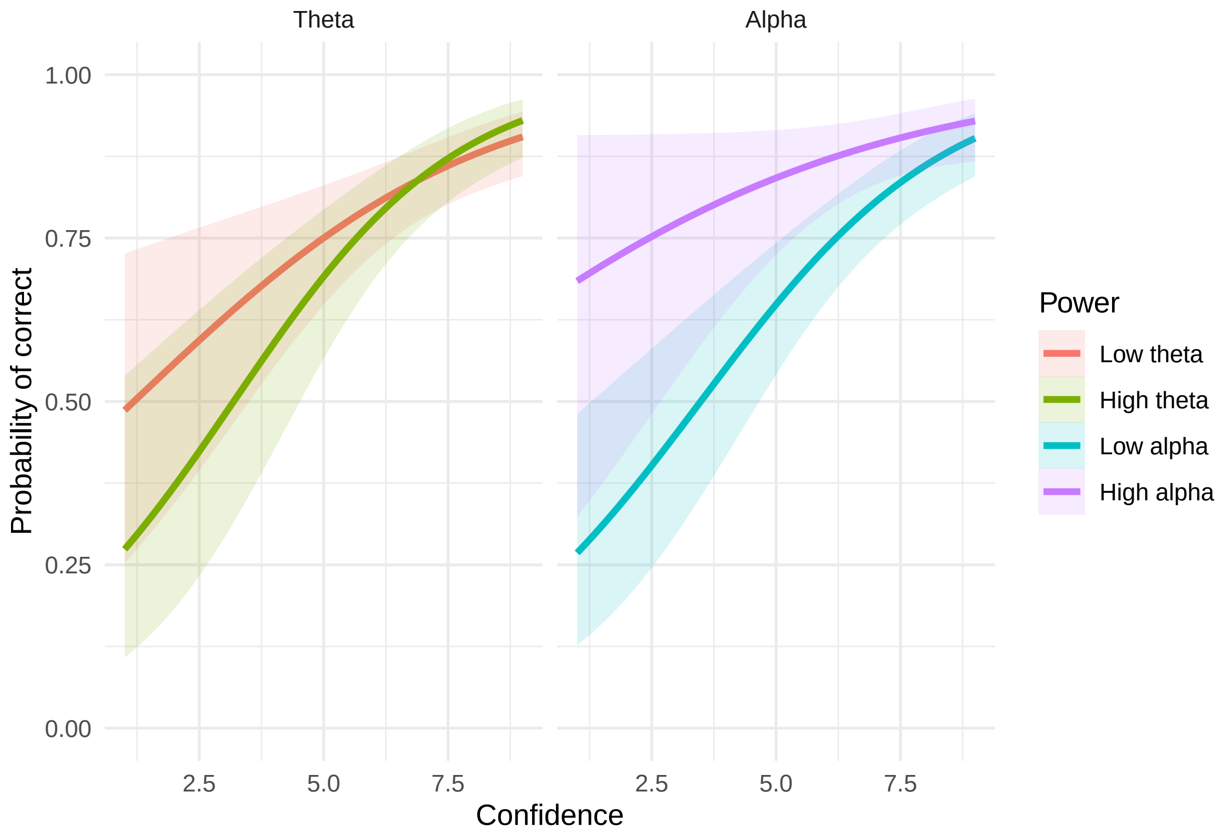
Plot med både alpha og theta

```
theta_data <- data.frame(
  confidence = full_data$confidence.x,
  correct = full_data$correct.x,
  group = full_data$theta_group,
  band = "Theta"
)

alpha_data <- data.frame(
  confidence = full_data$confidence.x,
```

```r
  correct = full_data$correct.x,
  group = full_data$alpha_group,
  band = "Alpha"
)


plot_data <- rbind(theta_data, alpha_data)

plot_data$band <- factor(
  plot_data$band,
  levels = c("Theta", "Alpha")
)

ggplot(plot_data,
       aes(x = confidence,
           y = correct,
           color = group,
           fill = group)) +

  geom_smooth(
    method = "glm",
    method.args = list(family = "binomial"),
    alpha = 0.15,
    size = 1.2
  ) +

  facet_wrap(~ band) +

  scale_y_continuous(limits = c(0,1)) +

  labs(
    x = "Confidence",
    y = "Probability of correct",
    color = "Power",
    fill = "Power"
  ) +

  theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
sd(full_data$theta_power)
```

```
## [1] 2.653983e-10
```

```r
sd(full_data$alpha_power)
```

```
## [1] 3.5751e-11
```

Analyse af spørgeskema:

```r
#Global Beliefs
belief_data <- read_csv2("GlobalBeliefs.csv",
                         col_types = cols(participant = col_character()))
```

```
## i Using "','" as decimal and "'.'" as grouping mark. Use `read_delim()` for more control.
```

```r
#names(belief_data)

belief_data <- belief_data %>%
  rowwise() %>%
  mutate(global_belief = mean(c_across(Q1:Q20), na.rm = TRUE)) %>%
  ungroup()

#Sanity Check
summary(belief_data$global_belief)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.400   1.400   1.475   1.675   1.750   2.350
```

```r
belief_data
```

```
## # A tibble: 4 x 22
##   participant    Q1    Q2    Q3    Q4    Q5    Q6    Q7    Q8    Q9   Q10   Q11
##   <chr>       <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 007             1     1     4     2     1     1     1     1     3     1     3
## 2 0606            1     1     4     1     1     2     1     1     2     1     1
## 3 69              1     1     4     2     1     1     1     1     3     1     3
## 4 1738            2     4     5     1     1     3     1     4     5     1     1
## # i 10 more variables: Q12 <dbl>, Q13 <dbl>, Q14 <dbl>, Q15 <dbl>, Q16 <dbl>,
## #   Q17 <dbl>, Q18 <dbl>, Q19 <dbl>, Q20 <dbl>, global_belief <dbl>
```

```r
nrow(belief_data)
```

```
## [1] 4
```

```r
belief_data %>% select(participant, global_belief)
```

```
## # A tibble: 4 x 2
##   participant global_belief
##   <chr>               <dbl>
## 1 007                  1.4
## 2 0606                 1.55
## 3 69                   1.4
## 4 1738                 2.35
```

```r
#normalize participant ID
meta_results <- meta_results %>%
  mutate(participant = as.character(as.integer(participant)))

belief_data <- belief_data %>%
  mutate(participant = as.character(as.integer(participant)))

belief_data_clean <- belief_data %>%
  select(participant, global_belief)

#combining
combined <- left_join(
  meta_results,
  belief_data_clean,
  by = "participant"
)

#inverter
#combined <- combined %>%
#  mutate(global_belief_flipped = 6 - global_belief)
combined$global_belief <- 6 - combined$global_belief
combined
```

```
##   model participant   dprime           c     metaD     Ratio global_belief
## 1    ML           7 2.168294  0.56605738 0.9222652 0.4253413          4.60
## 2    ML           8 2.256872  0.33679759 1.7766476 0.7872167            NA
## 3    ML         606 1.621542  0.01913239 2.1852962 1.3476655          4.45
## 4    ML        1738 1.788810  0.10276640 1.3275725 0.7421540          3.65
## 5    ML          69 1.378559 -0.26358385 1.0047329 0.7288284          4.60
```

```r
names(combined)
```

```
## [1] "model"         "participant"   "dprime"        "c"
## [5] "metaD"         "Ratio"         "global_belief"
```

```r
belief_data$participant
```

```
## [1] "7"    "606"  "69"   "1738"
```

```r
meta_results$participant
```

```
## [1] "7"    "8"    "606"  "1738" "69"
```

Lets a go

```r
#H5
cor.test(combined$Ratio, combined$global_belief)
```

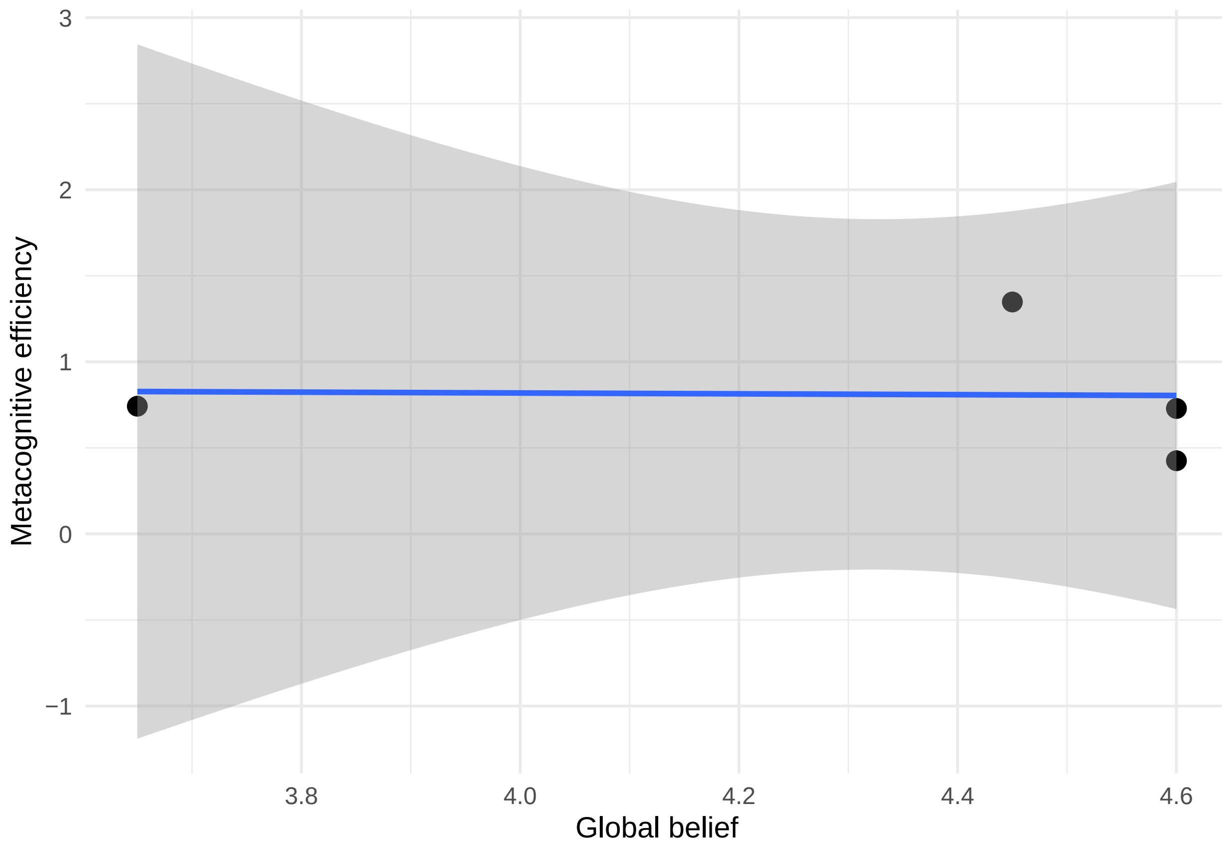
```
##
##  Pearson's product-moment correlation
##
## data:  combined$Ratio and combined$global_belief
## t = -0.040441, df = 2, p-value = 0.9714
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.9632101  0.9588442
## sample estimates:
##        cor
## -0.0285843
```

```r
ggplot(combined, aes(x = global_belief, y = Ratio)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    x = "Global belief",
    y = "Metacognitive efficiency"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



H4

```
# 1. Clean global belief data
belief_data_clean <- belief_data %>%
  mutate(
    participant = as.character(as.integer(participant)),  # fix IDs
    global_belief_flipped = 6 - global_belief             # flip scale (1-5 → 5-1)
  ) %>%
  select(participant, global_belief, global_belief_flipped)

# 2. Compute mean confidence per participant
confidence_summary <- data %>%
  mutate(participant = as.character(as.integer(participant))) %>%  # fix IDs
  group_by(participant) %>%
  summarise(mean_confidence = mean(confidence, na.rm = TRUE))

# 3. Merge datasets
conf_belief <- left_join(
  confidence_summary,
  belief_data_clean,
  by = "participant"
)
```

```r
# 4. Sanity check (IMPORTANT)
conf_belief
```

```
## # A tibble: 5 x 4
##   participant mean_confidence global_belief global_belief_flipped
##   <chr>                 <dbl>         <dbl>                 <dbl>
## 1 1738                   7.05          2.35                  3.65
## 2 606                    6.88          1.55                  4.45
## 3 69                     4.86          1.4                   4.6
## 4 7                      7.57          1.4                   4.6
## 5 8                      7.52         NA                    NA
```

```r
nrow(conf_belief)
```

```
## [1] 5
```

```r
# 5. Correlation test
cor.test(conf_belief$mean_confidence,
         conf_belief$global_belief_flipped)
```

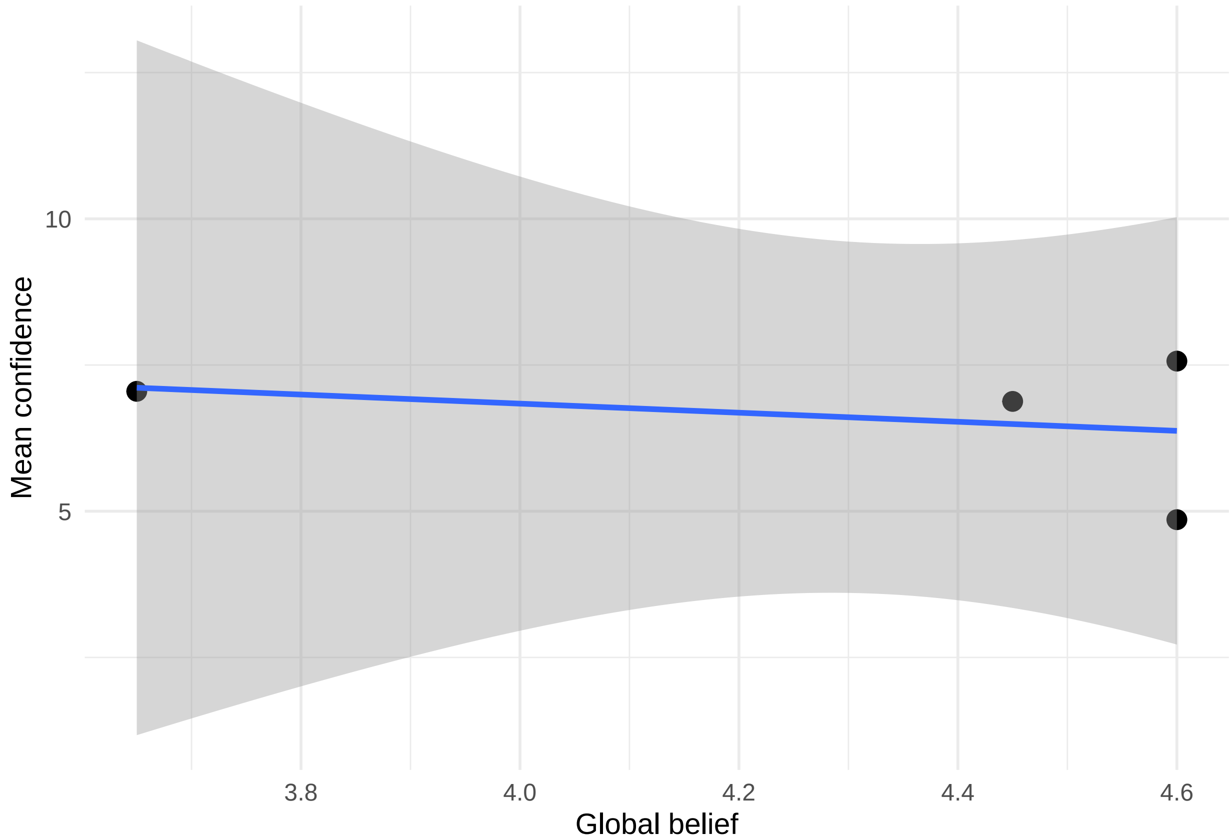
```
##
##  Pearson's product-moment correlation
##
## data:  conf_belief$mean_confidence and conf_belief$global_belief_flipped
## t = -0.43933, df = 2, p-value = 0.7033
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.9787035  0.9294181
## sample estimates:
##        cor
## -0.2966694
```

```r
# 6. Plot
ggplot(conf_belief, aes(x = global_belief_flipped, y = mean_confidence)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    x = "Global belief",
    y = "Mean confidence"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```
library(dplyr)
library(tidyr)

angle_conf_table <- data %>%
  #filter(participant != "69") %>%
  mutate(
    confidence_group = case_when(
      confidence <= 3 ~ "Low confidence",
      confidence <= 6 ~ "Medium confidence",
      TRUE ~ "High confidence"
    ),
    confidence_group = factor(
      confidence_group,
      levels = c("Low confidence", "Medium confidence", "High confidence")
    ),
    angle = factor(angle, levels = c("FL", "HL", "S"))
  ) %>%
  group_by(angle, confidence_group) %>%
  summarise(
    count = n(),
    .groups = "drop"
  ) %>%
  pivot_wider(
    names_from = confidence_group,
    values_from = count,
    values_fill = 0
```

```
  )

angle_conf_table
```

```
## # A tibble: 3 x 4
##   angle `Low confidence` `Medium confidence` `High confidence`
##   <fct>            <int>               <int>             <int>
## 1 FL                  53                  89               157
## 2 HL                  43                  77               181
## 3 S                   28                  43               229
```

```
angle_accuracy <- data %>%

  group_by(angle) %>%

  summarise(
    correct_trials = sum(response == correct_answer, na.rm = TRUE),
    total_trials = n(),
    accuracy = round(mean(response == correct_answer, na.rm = TRUE), 3),
    .groups = "drop"
  )

angle_accuracy
```

```
## # A tibble: 3 x 4
##   angle correct_trials total_trials accuracy
##   <chr>          <int>        <int>    <dbl>
## 1 FL               213          299    0.712
## 2 HL               246          301    0.817
## 3 S                270          300    0.9
```